

Assignment - 1.5

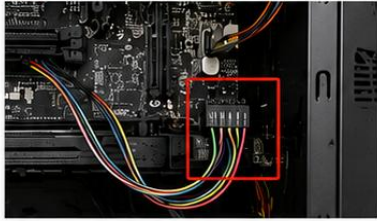
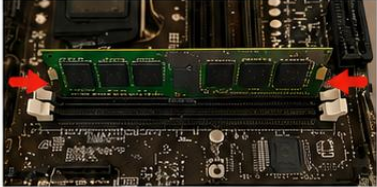
Q - 1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Component	Short Description
Motherboard	The main circuit board that connects all computer parts together.
CPU (Processor)	The brain of the computer that performs all calculations and processing.
CPU Cooler/Fan	Keeps the CPU cool to prevent overheating.
RAM (Memory)	Stores temporary data so programs run faster.
Hard Disk Drive (HDD)	Stores the operating system, software, and files permanently.
Solid State Drive (SSD)	A faster storage device than an HDD for quicker boot and loading times.
Power Supply Unit (PSU)	Supplies electrical power to all computer components.

Component	Short Description
Graphics Card (GPU)	Processes graphics and videos for the monitor.
Cooling Fans	Improve airflow inside the cabinet and keep components cool.
Optical Drive (if available)	Reads and writes CDs or DVDs.
Expansion Cards	Add extra features like Wi-Fi, sound, or additional ports.
SATA/Data Cables	Connect storage devices to the motherboard for data transfer.
Power Cables	Carry power from the PSU to the motherboard and other components.
CMOS Battery	Keeps the system clock running and saves BIOS settings when the PC is off.
Cabinet (Case)	Holds and protects all the internal computer components.

Q - 2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

DESKTOP PC DISASSEMBLY GUIDE

1 Unplug All Cables  Turn off the PC and unplug the power cable and all other connected cables.	2 Remove Side Panel  Remove the screws at the back and slide the side panel off.	3 Disconnect Front Panel Cables  Disconnect the front panel connectors (Power SW, Reset SW, HDD LED, etc.) from the motherboard.
4 Disconnect Power Cables  Disconnect the 24-pin motherboard power cable and the 4/8-pin CPU power cable.	5 Disconnect Data Cables  Disconnect SATA data cables and power cables from HDD/SSD and other drives.	6 Remove RAM  Push the clips on both sides of the RAM stick outward and lift the RAM stick out.
7 Remove Expansion Cards (GPU, etc.)  Remove the screws holding the card and gently pull it out from the PCIe slot.	8 Remove Storage Drives  Remove screws or unlock tray and slide out the HDD/SSD.	9 Remove CPU Cooler  Unplug the CPU fan cable. Remove the screws/pins and lift the cooler off the CPU.
10 Remove Motherboard  Remove all screws holding the motherboard and carefully lift it out of the case.	11 Clean Components (Optional)  Use a blower or soft brush to clean dust from components.	12 All Components Removed  All major components are now removed and the PC is fully disassembled.

Q - 3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

PC Reassembly Steps :

1. Install the Motherboard

- Place the motherboard in the cabinet and tighten the screws.
- **Why first:** It is the main board where all other components connect.

2. Install the CPU

- Carefully place the CPU into the CPU socket and lock it.
- **Why next:** The CPU must be installed before the cooler.

3. Install the CPU Cooler

- Attach the cooler and connect its power cable to the CPU_FAN header.
- **Why:** Keeps the CPU cool during operation.

4. Install the RAM

- Insert the RAM sticks into the correct memory slots until they click.
- **Why:** The system needs RAM to run programs.

5. Install the HDD/SSD

- Secure the storage drive and connect the SATA data and power cables.
- **Why:** Stores the operating system and files.

6. Connect the Power Supply (PSU)

- Connect the 24-pin motherboard cable and 8-pin CPU power cable.
- **Why:** Supplies power to all components.

7. Connect Front Panel Cables

- Connect the power switch, reset switch, USB, audio, and LED cables.
- **Why:** Allows the front buttons and ports to work.

8. Install the Graphics Card (if used)

- Insert the graphics card into the PCIe slot and connect PCIe power cables if required.
- **Why:** Provides video output and better graphics performance.

9. Check All Connections

- Make sure every cable is connected securely and nothing is loose.
- **Why:** Prevents startup problems.

10. Close the Cabinet and Test the PC

- Replace the side panel, plug in the power cable, and turn on the PC.
- **Why:** Confirms that the PC has been reassembled correctly.

Q - 4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Item	Why It Is Important
Turn off the PC and unplug the power	Prevents electric shock and damage to parts.
Wear an anti-static wrist strap	Protects components from static electricity.
Work on a clean, flat table	Keeps parts safe and organized.
Keep liquids away	Prevents damage from spills.
Use a Phillips screwdriver	Used to remove and tighten screws.
Keep a small container for screws	Stops screws from getting lost.
Handle parts by the edges	Prevents damage to electronic components.
Disconnect cables gently	Avoids breaking cables and connectors.
Take photos before removing parts	Makes reassembly easier.
Read the PC manual if available	Helps identify parts and connections correctly.